Ayush Singh

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Education

University of Illinois Urbana-Champaign Champaign, IL
B.S. in Computer Science + Linguistics *Expected May 2027*
GPA: 3.85/4.00

Relevant Coursework: Data Structures and Algorithms, Machine Learning, Computational Linguistics, Computer Architecture, System Programming, Deep Generative Models, Distributed Systems, ML System

Experience

Member of Technical Staff Intern June 2026 – Present
Inflection AI *Palo Alto, CA*

- Built the GRPO post-training stack for a production TTS model—custom training environments for task specification and rollout collection, plus a three-tier reward function combining catastrophic-failure hard gates, a weighted composite over ASR word-error-rate, semantic-embedding similarity, and multi-model MOS predictors, and multiplier gates for noisy detectors; shipped a production-ready fine-tuned checkpoint.
- Found that a MOS predictor inside the reward function applied unseeded random cropping, corrupting completion rankings *within* each GRPO sample group: across 13,104 production rollout groups the reward-selected best-of-K flipped 16–47% of the time. Removed it in a live A/B pilot, cutting hard-floor violations 30% relative (2.47%→1.74%) at +1.0% mean reward.
- Cut a word-level timestamp head's "stuck word" error rate from 29.3% to 22.0% (3-seed ensemble) via a LoRA layer/rank search informed by a hidden-state-collapse study across all 24 decoder layers, plus a monotonic alignment curriculum on an already-converged checkpoint; added Viterbi decoding raising word coverage 90.3%→97.1% at no measured accuracy cost.

Research Fellow February 2026 – Present
SPAR AI *Berkeley, CA*

- Developing **SelectiveLoRA**, applying low-rank adaptation (LoRA) adapters to interpretability-selected layers rather than uniformly, achieving 37–50% parameter reduction on Qwen2.5-1.5B; benchmarking across model families (Llama, Qwen, Gemma) and task types (math, code, commonsense, factual).
- Analyzing fine-tuning dynamics using sparse autoencoders (SAEs) to track feature birth, death, and drift during domain adaptation, probing whether early feature geometry changes can predict catastrophic forgetting.

Research Assistant January 2026 – Present
University of Illinois Urbana-Champaign *Champaign, IL*

- Co-author paper on verifier reliability in execution-based evaluation **accepted at INLG 2026**; advised by Aleksandre Maskharashvili.
- Diagnosed systematic failure modes in LLM-generated pandas verifiers—majority quantifiers evaluated under universal truth conditions, verdicts contradicting their own zero-violation justifications—establishing that 89% of checker–human disagreements were false negatives and that reported accuracy is a conservative lower bound.
- Quantified verifier reliability across model scales, showing checker fidelity degrades sharply below 70B (99% agreement for GPT-OSS-120B vs. 81% for Llama-3.1-8B) and that execution-based evaluation is only trustworthy when the code-generating model clears a capability threshold.

Member of Technical Staff Intern June 2025 – August 2025
Inflection AI *Palo Alto, CA*

- Contributed to PyTorch's Torchtune library by implementing step-based and async checkpointing, enabling up to 10x faster model saving during fine-tuning; merged via PR #2869.
- Improved internal data flywheel with parallel execution, increasing preference data synthesis by 50% across 112K scenarios; adapted MT-Bench and SAGE Bench for EQ-specific evaluation.

Technical Skills

Languages: Python, Java, C/C++, Bash, YAML
ML/Research: PyTorch, Torchtune, TRL, Transformers (HuggingFace), Scikit-learn, XGBoost, Pandas, NumPy, RLHF, Post-training, LLM Evaluation
Interpretability & Alignment: Sparse Autoencoders (SAEs), LoRA, DPO, GRPO, Linear Probing, Reward Modeling
Infrastructure: SLURM, vLLM, Ray, Weights & Biases, Git/GitHub, CI/CD, Docker, Azure